BATCH 18

Entrained-Flow Coal Gasification

Reduced-order simulation of the BYU atmospheric reactor

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Selected paper: Mularski and Modlinski (2021), Energies 14(6), 1729

ME40249 | Gasification and Reforming Processes

The assigned case

Atmospheric oxygen-blown BYU gasifier

A 0.20 m diameter x 1.80 m reactor processes 23.88 kg/h of Utah bituminous coal. The selected paper reports two coaxial feeds and a Rosin-Rammler coal-size distribution.

Feed	Rate	Inlet	Composition
Primary	26.24 kg/h	400 K	85% O2, 12.6% Ar, 2.4% H2O
Secondary	6.62 kg/h	473 K	Steam
Coal	23.88 kg/h	-	Utah bituminous, 1-85 um

Paper: Mularski and Modlinski (2021), Table 3. Particle distribution: mean scale 36 um, spread 1.033.

Utah coal composition

The source reports proximate analysis as received and ultimate analysis on a dry ash free basis. Oxygen is calculated by difference.

Proximate	wt%
Moisture	2.40
Volatile matter	45.60
Fixed carbon	43.70
Ash	8.30

Ultimate, DAF	wt%
Carbon	77.60
Hydrogen	6.56
Nitrogen	1.42
Sulfur	0.55
Oxygen by difference	13.87

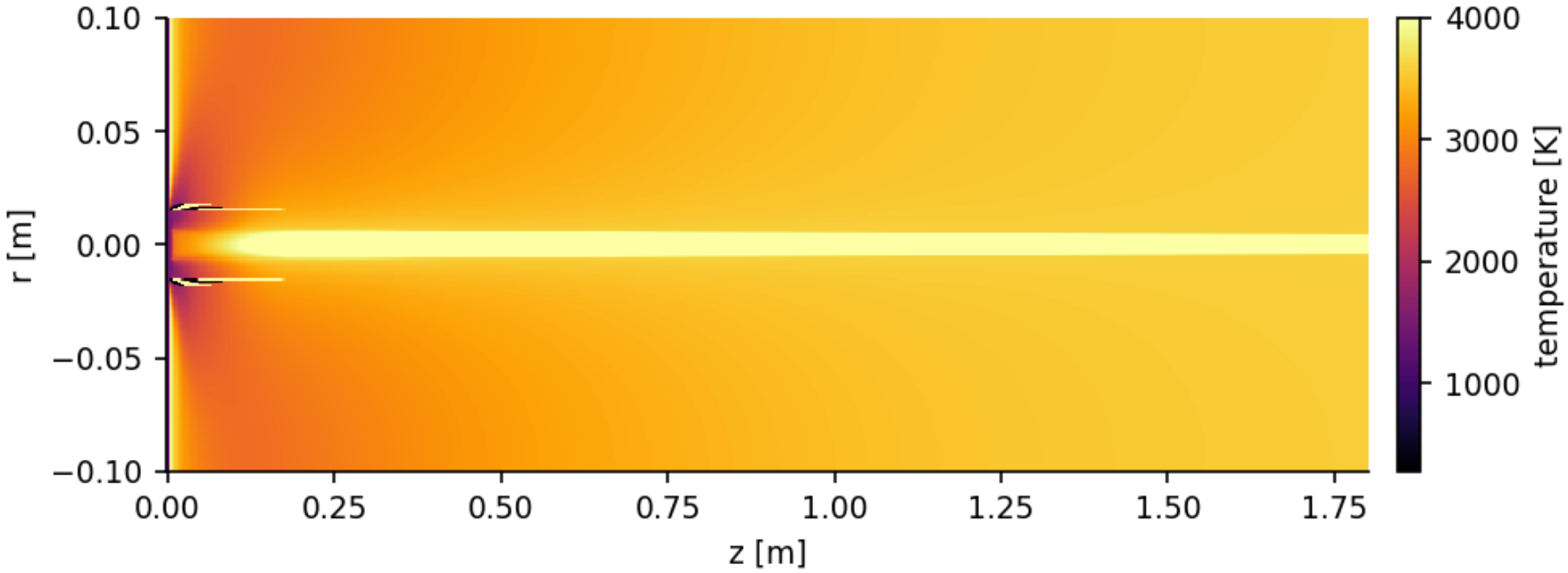
HHV input: 33.52 MJ/kg DAF estimated with the Channiwala-Parikh correlation. Particle density is a provisional assumption.

Reduced model and reference CFD

Model feature	Reference CFD	Batch 18 model
Flow	2-D axisymmetric turbulence model	Radial stream tubes, no momentum equation
Particles	Trajectories and dispersion	Coal stays in its inlet annulus
Heat	Radiation and wall terms	Adiabatic wall, no radiation
Coal chemistry	C2SM/FG-DVC; char kinetics fitted to CBK/E and CBK/G	C2SM Table 5 and BYU-fitted char parameters from Table 6; reduced gas chemistry
Mixing	Turbulence-chemistry interaction	Assumed radial diffusion, operator split

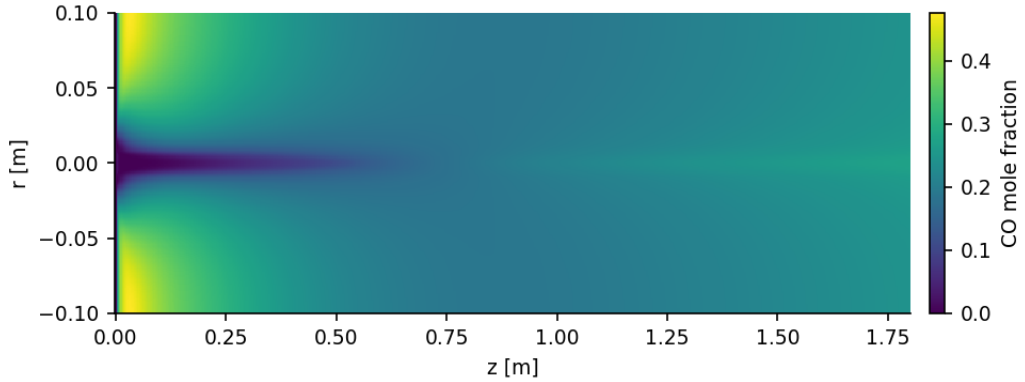
One solver workflow applies axial chemistry across radial annuli with species and heat exchange. The 1-D lanes describe the axial chemistry; the radial calculation resolves cross-stream variation. These describe spatial detail, not project phases. Momentum, particle trajectories and radiation are omitted. The 1400 K ignition patch adds 19.35 kW (9.75% of coal HHV input).

Temperature field

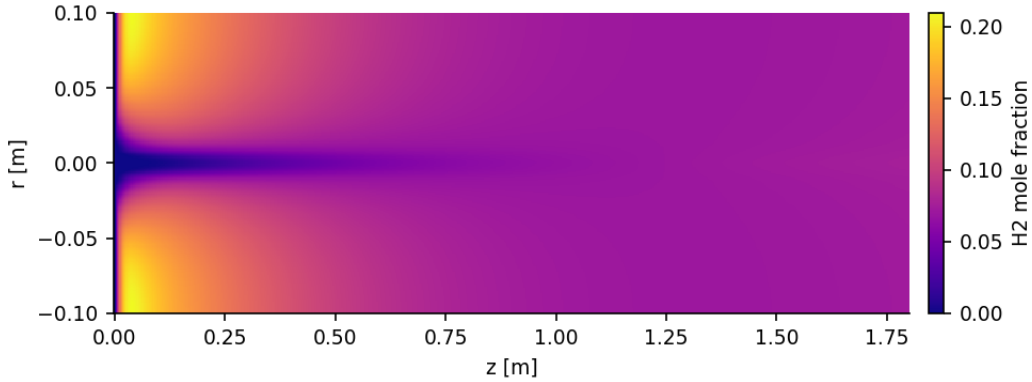


Peak reported at 4000 K, the solver's temperature-recovery ceiling. The true peak is unresolved. Outlet area-average temperature is 3556.7 K. The 1400 K hot-start patch supplies 9.75% of coal HHV input.

Species contours: CO and H2



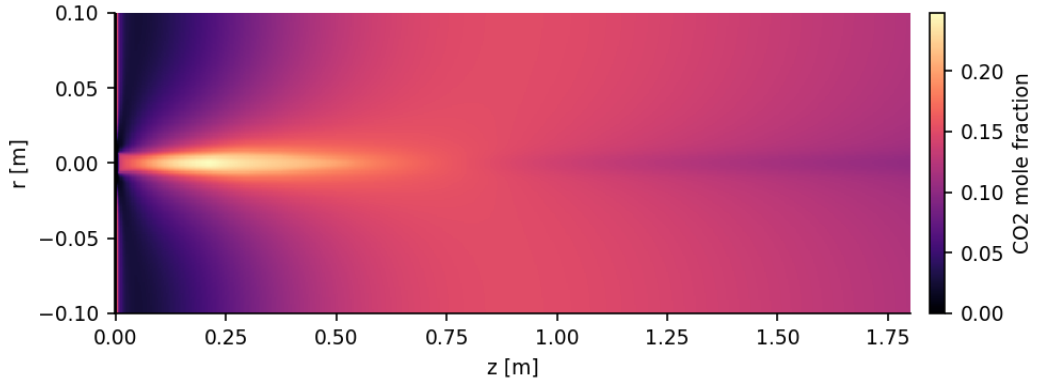
CO



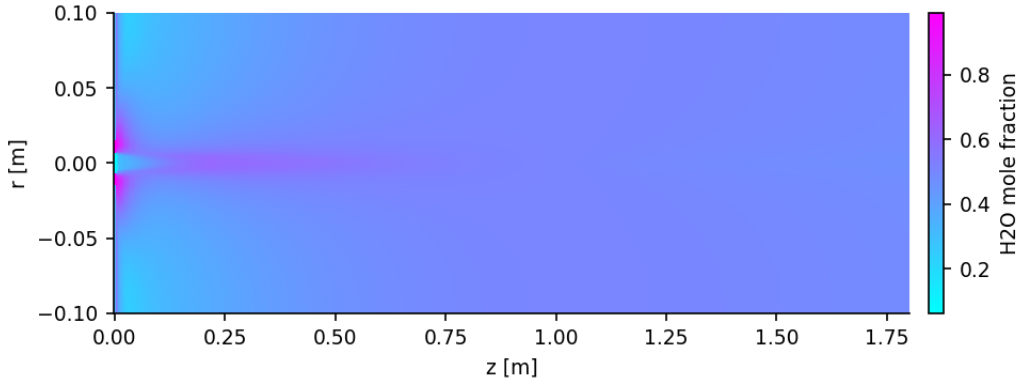
H2

Local mole fractions in the radial-annulus model.

Species contours: CO2 and H2O



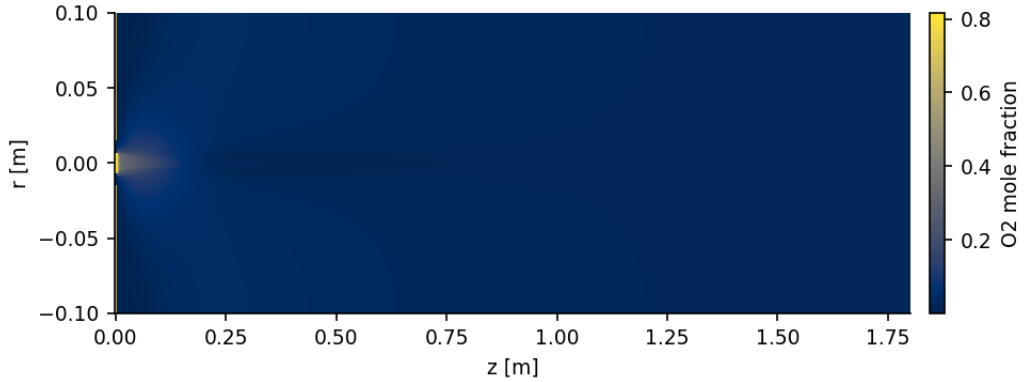
CO2



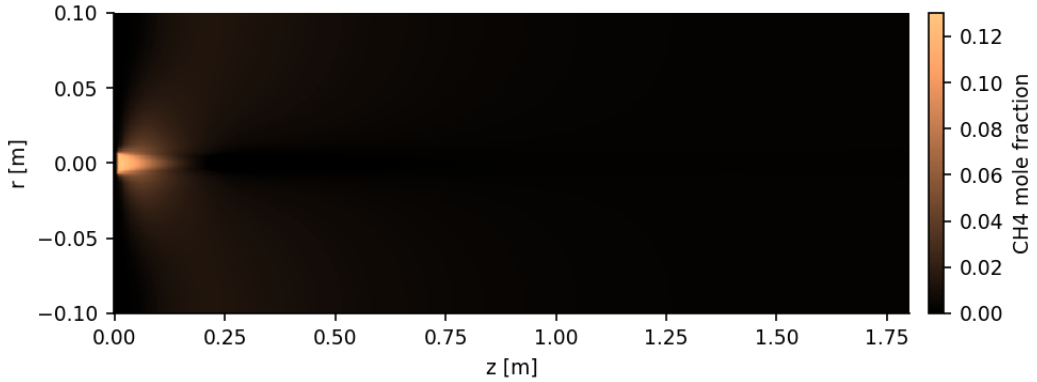
H2O

Water measurements carry about +/-14% uncertainty in the source study.

Species contours: O2 and CH4



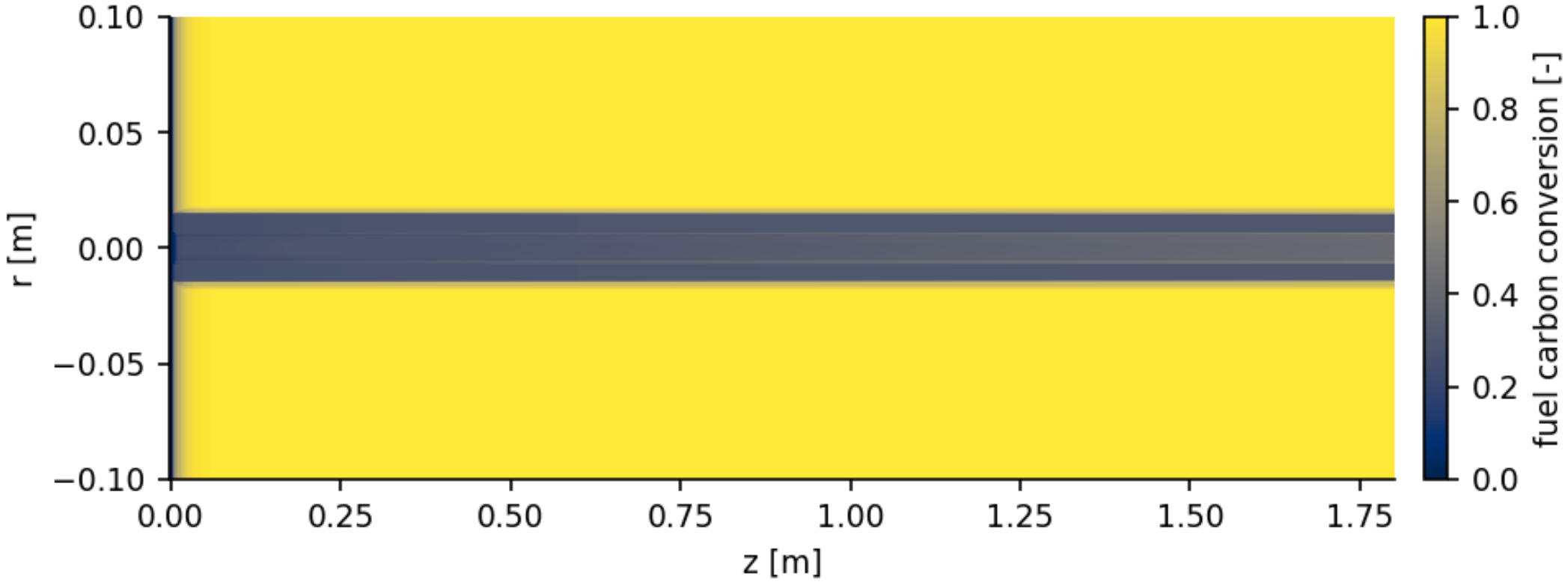
O2



CH4

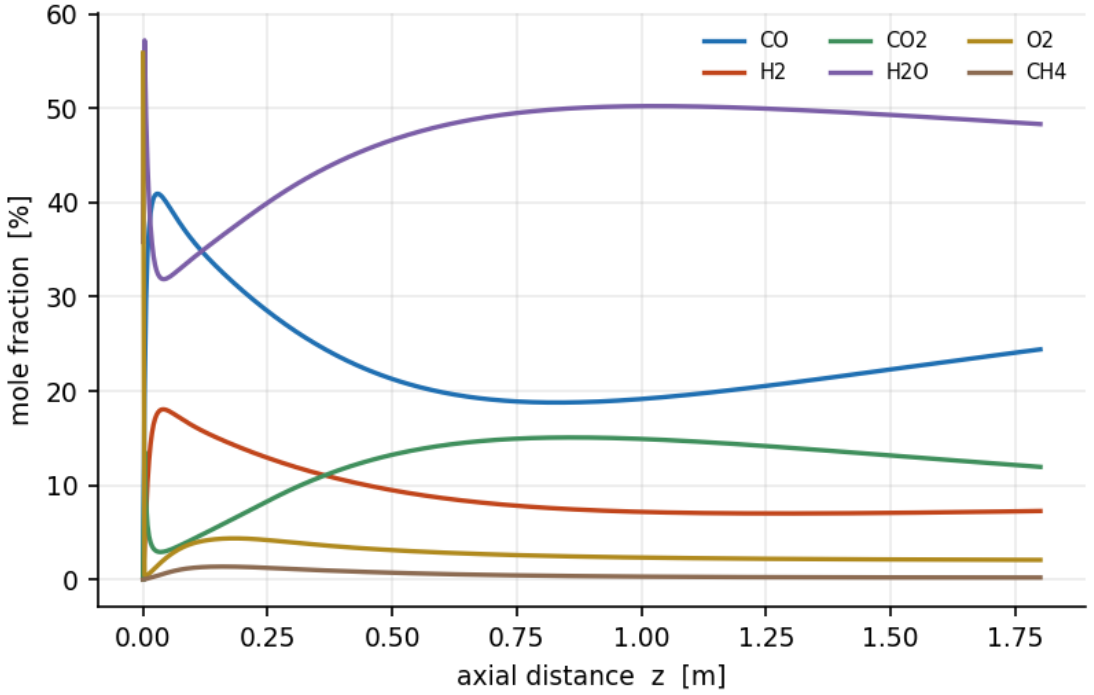
The model leaves 3.88 vol% dry O2 at the outlet.

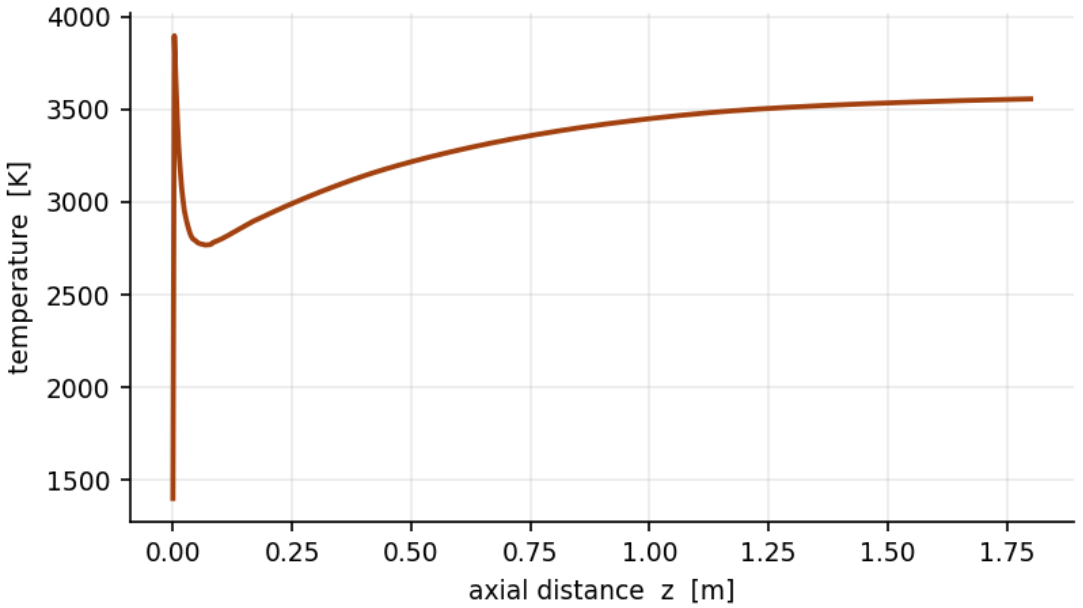
Fuel-carbon conversion field



The field uses the remaining char and raw coal in each annulus. Its near-uniform high values are another sign that this chemistry saturates locally.

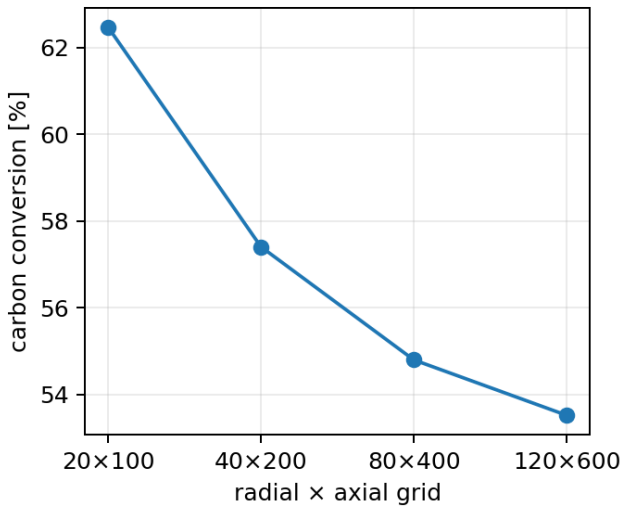
Axial species and temperature profiles

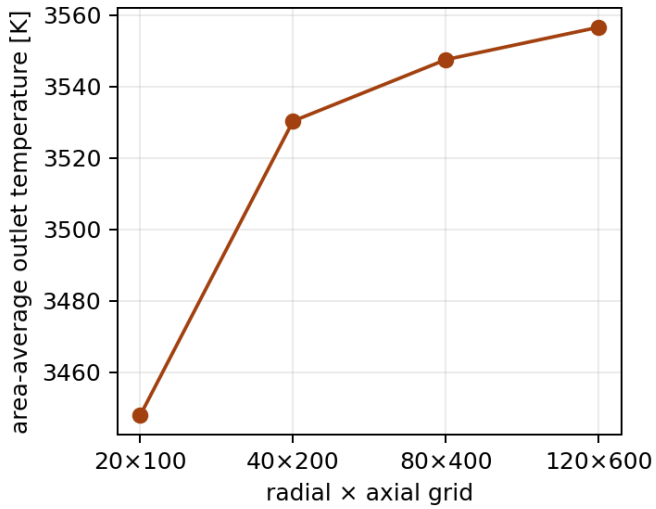




Both plots show cross-sectional area averages, not centerline probe data.

Numerical verification





120 x 600 grid

Element closure

Maximum absolute C/H/O/N/Ar balance error: 0.0000021%

80 x 400 to 120 x 600

Outlet T change: 9.1 K

Conversion change: -1.28 percentage points

Main dry gas changes: 0.65-0.80 percentage points

The peak is capped on every grid, so peak-temperature convergence is unknown. Grid refinement does not establish physical accuracy.

Comparison with BYU experiment

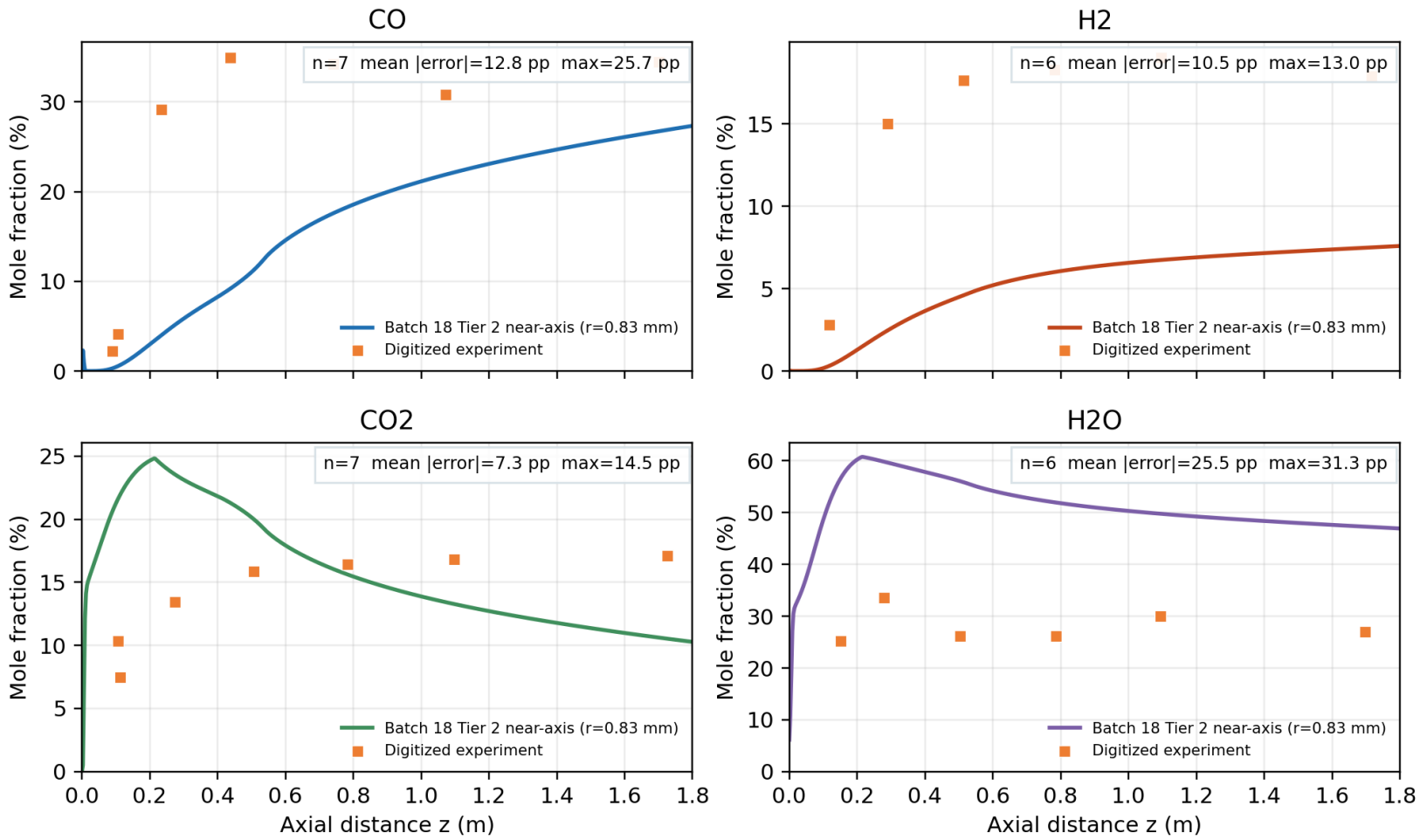
Result	Batch 18	Experiment	Difference
CO, dry inert-free	58.66%	51.1%	+7.56 pp
H2, dry inert-free	16.68%	24.7%	-8.02 pp
CO2, dry inert-free	24.21%	23.6%	+0.61 pp
CH4, dry inert-free	0.45%	0.3%	+0.15 pp
Carbon conversion	53.52%	82%	-28.48 pp

Interpretation

Element balance closes, but the model underpredicts conversion and H2 and leaves oxygen in the outlet. The next two slides compare near-axis and radial profiles with digitized experimental markers; the paper's Table 8 errors describe its own CFD.

Near-axis comparison with centerline data

Near-axis model profile versus centerline markers from Figure 9



Mean / max absolute error (pp)

Species Batch 18 Paper CFD

CO 12.81 / 25.72 4.82 / 18.48

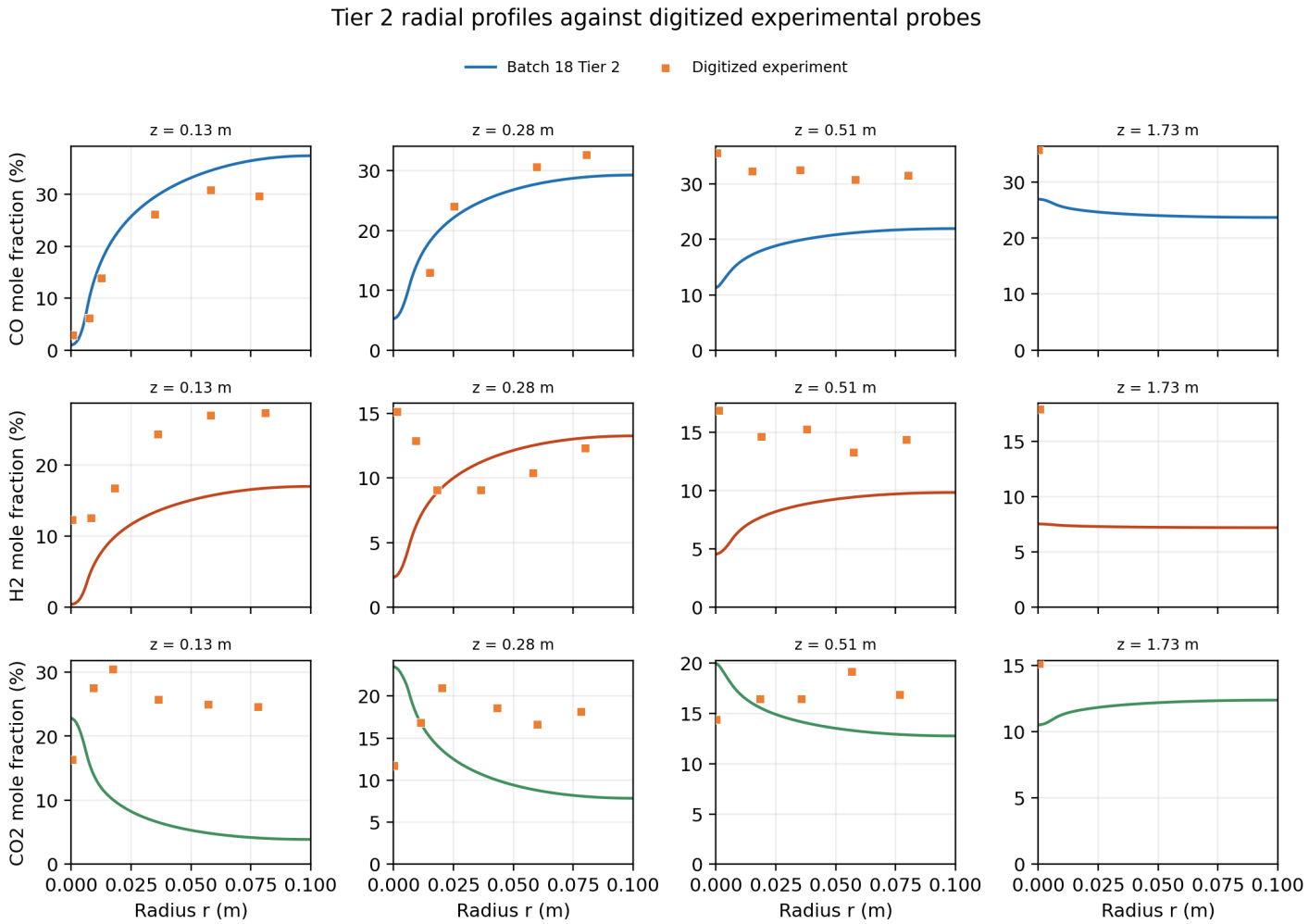
H2 10.50 / 13.03 5.47 / 7.95

CO2 7.27 / 14.46 4.72 / 10.15

H2O 25.54 / 31.34 9.58 / 20.22

Digitized experimental markers from Mularski & Modlinski (2021), Fig. 9. Errors are absolute composition differences in percentage points. H2O carries about +/-14% measurement uncertainty and is excluded from the paper's quality score.

Radial comparison with experimental probes



Pooled mean / max error (pp)

CO 8.54 / 24.22 (n=34)

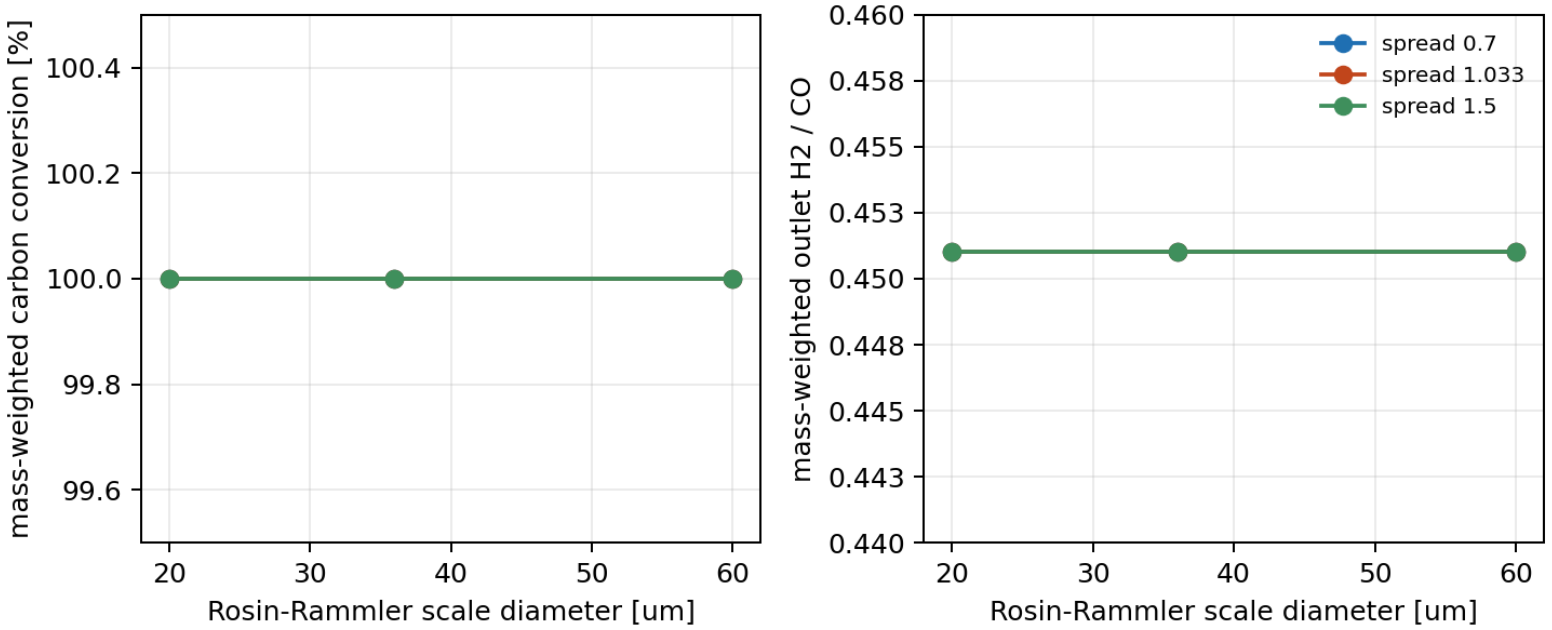
H2 7.02 / 15.33 (n=36)

CO2 6.74 / 20.51 (n=35)

All eight stations are scored; four are plotted. Orange markers were digitized from Figures 12-14 of Mularski & Modlinski (2021). Estimated digitization uncertainty: +/-0.001 m radius and +/-0.5 pp composition.

H2O is excluded because the paper derives it by hydrogen balance. Errors are absolute mole-fraction differences in percentage points. The comparison shows meaningful local gaps; the reduced model does not reproduce all radial structure.

Particle-size distribution diagnostic



Six mass-weighted Rosin-Rammler bins; scale diameter 20, 36, 60 um and spread 0.7, 1.033, 1.5. The cross-section-averaged calculation is flat: 100% conversion, 71.65% CGE and H2/CO = 0.451 in all cases.

This diagnoses complete-burnout saturation in the cross-section-averaged calculation. It does not demonstrate that particle size has no effect. The reference study reports 83% conversion at 36 um mono and 77% for the full distribution, versus 82% measured.

Conclusions

The 120 x 600 model conserves tracked elements to about 0.0000021%.

Outlet temperature is 3556.7 K; peak temperature reaches the 4000 K ceiling. The hot-start patch adds 9.75% of coal HHV input.

Carbon conversion is 53.52%, 28.48 percentage points below the 82% experiment.

Dry inert-free H2 is 16.68% versus 24.7% measured, and dry outlet O2 remains at 3.88%.

Pooled radial mean errors are 8.54 pp for CO, 7.02 pp for H2 and 6.74 pp for CO2; substantial local gaps remain.

The current particle-size sweep is inconclusive because the cross-section-averaged calculation saturates at 100% conversion.

Administrative check: roster and case are recorded; operating-condition uniqueness should be confirmed with the instructor.